Countersign

Catch the scam while it's still a call.

Team losers · RAKSHAM 2026 finale · Track 1

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the code and every result

Mummy's phone rings. A man says he's from the CBI: a parcel in her name had drugs in it, she must stay on the line and tell no one. An hour later she moves her savings to his "safe account". Digital-arrest, fake-KYC and "your son had an accident" calls all run like this, and **by the time the call ends, the money is gone.**

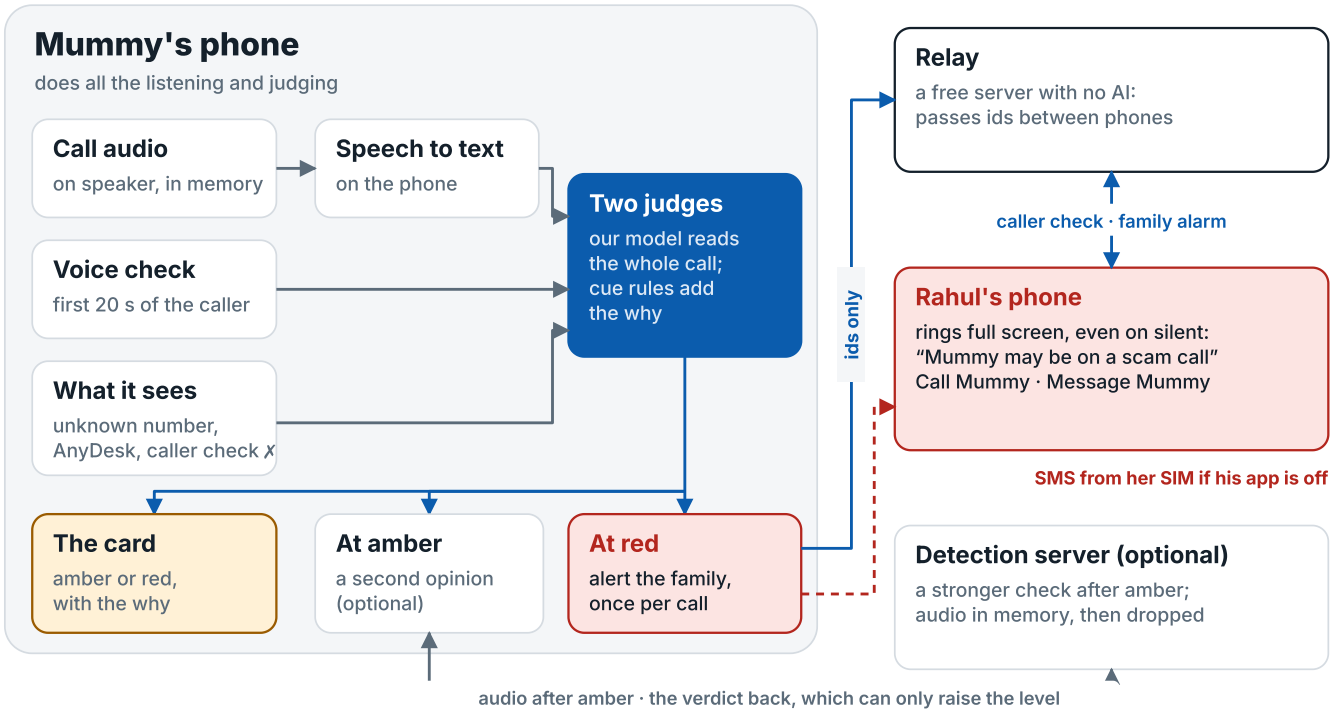
What Countersign does

1 Is it really Rahul?
When a caller claims to be family, the app asks Rahul's own phone whether it is on this call. A cloned voice or a faked number fails, because Rahul's phone never placed the call.

2 It listens with her, on the phone
On speaker, the phone turns the call into text itself and our trained model judges what the caller is doing. The card turns amber and says why: "says police, court or bank · asks you to keep it secret".

3 At red, the family hears
Rahul's phone rings full screen, even on silent. If his app is off, Mummy's phone texts him; if nobody acts within a minute, it calls him.

4 A mistake costs two taps
"I know this caller", then "Yes, it's fine": the warnings stop for this call and Rahul is told it was a false alarm.



2.3%
of genuine calls flagged
1,019 bank calls it never trained on (ICFD-31k). Our first, keyword version: 50.1%.

88%
of scam calls caught
4,499 held-out scam calls (ICFD-31k), at amber or red.

28/30
on our spot-check kit
10 calls, 10 messages, 10 voices; one command reruns it.

10/10
new scams caught
Written after the model was fixed; each call also tested over a phone line.

Every number: docs/RESULTS.md in the repository, reproducible on a laptop.

One call, step by step



1 It rings
Is it Rahul? His own phone is asked.

2 On speaker
Speech to text on the phone. Nothing is saved.

3 Amber
The scam named, and why.

4 Red
Two independent sources agree.

5 Rahul's phone
Rings over silent; SMS if his app is off; a call after 60 s.

Wrong? Two taps.

I know this caller
or "It's not a scam"

→ **Is this call fine?**
→ Yes, it's fine

→ **Warnings stop**
for this call

→ **Rahul sees**
"False alarm: Mummy says it's fine"

How it decides

Two judges read every call. Our model, fine-tuned on 31,000 Indian bank calls, reads the whole call so far and asks whether the caller is doing what scammers do. The cue rules spot the moves in English, Hinglish and Hindi, and give the reason in plain words. What the phone sees counts too: an unknown number, AnyDesk opened, a failed caller check.

Amber needs one of them. Red needs two independent sources, so one odd sentence never sets off the family alarm. During a call the level never goes down, unless she says it's not a scam. A computer-made voice can raise a call to amber, never to red: over a speakerphone that check makes too many mistakes to decide alone.

Privacy, built in

- The call becomes text on the phone. Audio and words are never saved or sent.
- The relay that links family phones sees ids only: who, and which kind of scam.
- Only a call already at amber can go to our server for a second opinion, and its audio stays in memory.
- For a confirmed scam the server keeps one 20-second encrypted clip, deleted after 30 days.

What it misses

- A scam read politely from a script in English can stay below amber.
- People talking *about* a scam can set it off.
- A genuine call in a computer-made voice can reach amber; the card says why.
- On the earpiece the phone hears mostly her, so the call has to be on speaker.

What's next

- A full-screen stop over payment apps. It is designed; today the call card and the home screen warn.
- A small language model on the phone as a third judge. The prompt is written and tested.
- A bank pilot: our server already runs the call-centre version, with app approval for risky requests.

Built on multilingual-e5-small (fine-tuned by us), Meta's Omnilingual speech-to-text, NII's AntiDeepfake voice model and the ICFD-31k dataset. It runs on a 6 GB phone after one 798 MB download. **Built here:** 66% of the product code was written at the finale.

Try it yourself. On the phone: Home → *Check a recording or message*, with your own file or SMS. On a laptop, the same verdicts for a whole folder: `python -m ml.eval.kit <folder>`